Envisioning an emission diet: Application of travel demand mechanisms to facilitate policy decision making

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ABSTRACT:
Emission reduction strategies are gaining attention as planning agencies work towards adherence to air quality conformity standards. Policymakers struggling to reduce Greenhouse Gases (GHG) must grapple with a growing number of travel demand policies. To consider any of these emerging demand mechanisms as a viable option to meet emission targets, planners and policymakers need tools to better understand the implications of such policies on travel behavior. In this paper we present an integrated multimodal travel demand and emission model of four policy strategies; presenting GHG and air pollutants reduction results at a very detailed level. Multiple policy outcomes are compared within a single modeling framework and study area. The results reveal that while no one demand mechanism is likely to result in emission reductions that meet policy-maker's goals; a first-best pricing strategy that incorporates marginal social costs is the most effective emission reduction mechanism. Implementing such a mechanism may offer total emission reductions of up to 24%. However, the efficacy of this strategy must be weighed against substantial difficulties of establishing efficient pricing, a costly implementation, and substantial negative impacts on non-highway facilities. Decision makers must select a mixture of pricing and land use strategies to achieve emission goals on all road facilities.

Key Words: emission reduction, greenhouse gas, multimodal travel demand, pricing,

1. INTRODUCTION

Vehicular emissions continue to grow with industrialization and automobile ownership. Commonly cited potential causes, among many, include excessive commuting distances; under-utilization of public transit; slow retirement of less efficient vehicles and the low cost of driving. Some or all of these factors influence travel behavior that produces excessive emissions. A multitude of schemes have been devised to reduce transportation sector related emissions. The policies developed to reduce emissions vary from supply constraints to pricing incentives, all aimed at producing more efficient travel patterns. A number of studies have proposed emission reduction strategies through a variety of approaches. Examples include:

- changes in fuel formulation *(Schmidt and Van Gerpen, 1996; He et al., 2003),*
- accelerated vehicle retirement *(Deysher and Pickrell, 1997; Dill, 2004),*
- changes in vehicle fleet mix *(Ichinohe and Endo, 2006)*
- Vehicle propulsion technology *(Sioshansi and Denholm, 2009),*
- ride share programs *(Skene and Hanslip, 1994),*
- urban form *(Anderson et al., 1996; Williams et al., 2000; Camagni et al., 2002),*
- commute efficiency *(Scott et al., 1997; Horner and O'Kelly 2007; Layman and Horner, 2010),*
- telecommuting *(Roth et al., 2008),*
- road pricing *(Beevers and Carslaw, 2005; Mishra and Welch, 2012),*
- network analysis mechanisms *(Chin, 1996).*

While these studies have added significant value with regard to individual performance of emission reduction strategies, simultaneous analysis of travel behavior and multiple emission reduction policy mechanisms is not prominent in literature. This paper addresses two of the major policy directions in the current effort to reduce vehicle emissions, which we refer to as emission reduction mechanisms. The first group of mechanisms influences traveler behavior by imposing a higher cost for travel so that minimal emissions are produced in the system. This group of pricing mechanisms includes an array of policy

options from direct emissions charging to changes in the gas tax. For the second sets of mechanisms emission reductions are achieved primarily through non-network based policies. Such strategies involve land use, commuting behavior and vehicle efficiency requirements.

Many of these mechanisms have been analyzed individually in the literature; however, the mechanisms are rarely modeled within an empirically calibrated travel demand model or examined to determine how they would perform in a realistic transportation environment. Many studies focus on a single application of a mobile emissions model. A much smaller number integrate an emission estimation tool directly into the highway assignment algorithm of the travel demand model. As a result, literature shows significant variation in the resulting emission reduction from each mechanism, with little agreement among scholars. One way the efficacy of these cost and behavioral responses can be measured is with the development of travel demand or behavioral models. Such models are capable of examining travel behavior of road users and their responses to impedances such as emission pricing, to achieve emission reductions.

This paper addresses travel behavior aspects of emission reductions, however beyond the scope of this paper are still important issues of equity, uses of revenue from pricing mechanisms and limitations of policy. By necessity we limit the discussion to the boundaries of behavioral influence based on a mix of empirical, projected and theoretical inputs; leaving other issues to future discussion. Rather, proffering a complex formulation we focus on commonly accepted methods to examine performance of each of the major emission reduction mechanisms.

The remainder of this paper is organized into five sections. The first section reviews literature on emission reduction mechanisms both from the supply side and the travel demand side. The second section presents the modeling framework and policy formulations of the proposed emission reduction mechanisms. The third section describes the case study area. Results of the modeling application are presented in the fourth section followed by a discussion, conclusions and suggestions for further research in the fifth section.

2. RELEVANT LITERATURE

Vehicle emission reduction has long been a debated topic in the literature. The importance of the subject grew in 1970 in the United States (US) with the passage of the Clean Air Act (*Clean Air Act of 1970*, 1970). This Act gave the Environmental Protection Agency (EPA) the authority to regulate motor vehicle emissions for the first time in the agency's history (Bolbach, 1974). Since the law's enactment, the literature has filled with a variety of theoretical methods aimed at reducing emissions to achieve policy goals. Topics addressing reductions in emission run the gamut of supply-side (vehicles, modes and road provision) and demand-side (pricing and efficiency) transportation management. This paper focuses on demand-side mechanisms.

The demand side slate of options offers a fairly robust set of tools. These mechanisms can be sub-categorized in a number of ways. We find two distinct demand-side policy directions in the literature. The first grouping of policies is aimed at reducing the demand for vehicle miles travelled through some type of pricing mechanism. The second category of demand-side mechanisms is composed of policies that seek efficiency through changes in land use locations and subsequent travel decisions.

The first category of demand-side emission reduction mechanisms works by increasing the price of travel. One of the more common methods found in the literature uses mechanisms either adapted from congestion reduction strategies or presents an attempt to jointly reduce emissions and congestion. However, congestion and emissions are two separate transport externalities that are often incorrectly conflated. Congestion is primarily a quasi-externality, that is, those that cause congestion (by driving their vehicle) generally pay for the congestion effects of their driving by suffering extended travel time as a result of their travel decision. However an unpaid portion of the congestion, the delay caused by the individual choice to drive, impacts the travel time of other drivers on the same link. Drivers give up time, in the form of opportunity costs, to drive on a congested road (Walters, 1961). Drivers stuck in traffic also pay with higher fuel consumption. The environmental impacts of driving, namely the production of ozone-creating and heat-trapping compounds into the atmosphere, are paid for in the form of harm to

breathing, obstruction of views and other effects predicted from climate change by those not necessarily directly involved in the creation of such effects. Thus the environmental impact from transportation is largely a negative externality (Yin and Lawphongpanich, 2006). The difference in effect of these two transportation phenomena (congestion and emissions) is such that the cure for either should be different. Conceptually the two may seem to be directly related, but the complexity of both occurrences often means that mitigating one can lead to an increase in the other (Nagurney, 2000).

A growing body of research has distinguished the two externalities, suggesting direct emissions-based pricing (Johansson-Stenman and Sterner, 1997; Mishra and Welch, 2012), while others institute congestion pricing and measure emissions benefits (Daniel and Bekka, 2000). These studies fall under the first-best theory in which pricing is meant to capture the marginal social costs of the externality. Other pricing approaches fall under the theory of the second-best which includes increases to vehicle or fuel taxation (Fullerton and West, 2002; Sterner et al., 1992), charges based on the number of vehicle miles traveled (Greene, 2011); charging for travel that occurs within a cordon (Beevers and Carslaw, 2005; Chin, 1996) and reductions through a combination of pricing mechanisms (Deakin et al (1996).

The second category of demand-side mechanisms, called non-network mechanisms, work to encourage more efficient travel by reorganizing urban form (Anderson et al., 1996; Camagni et al., 2002; Muniz and Galindo, 2005; Williams et al., 2000) for example, by encouraging a balance between residential and job location (Horner and O'Kelly, 2007; Horner, 2008, 2002; Layman and Horner, 2010; Scott et al., 1997). In 2004 the large Propolis report tested numerous transportation and land-use scenarios for eight European metropolitan areas using multiple modeling frameworks. The results of the study indicate a combination of scenarios might provide significant emission reductions, somewhere in the range of a 15-20% reduction in CO_2 emissions (Steadman et al., 2004). Other mechanisms attempt to reduce congestion and emissions with a reduction in the demand for light-duty vehicle (LDV) transport. This is accomplished through a list of measures specified from the 1990 Clean Air Act (US EPA, 1990). This category of policies may be faster to implement but is often criticized for producing very small

emission reductions. The set of tools, called transportation control measures, are mostly policy-based options to encourage programs that make more efficient use of LDVs during commute hours, increase transit mode share and promote non-motorized travel. Other policies encourage higher vehicle occupancy using ride share programs (Skene and Hanslip, 1994) with emission reductions of 3-5%. Finally, a long-debated strategy has been to replace home-based work travel with telecommuting; though many studies find only marginal emission reductions, in the range of 0.16-0.23% carbon dioxide equivalents (CO_{2e}) for all 4 million US telecommuters (Roth et al., 2008).

Other studies focused on developing more sophisticated travel demand tools, modeling highly disaggregate departure times and second-by-second highway assignment. These models fall within a class called activity-based models (Ben-Akiva and Bowman, 1998; Bowman and Ben-Akiva, 2001; Kitamura, 1996, 1988) where household or individual activity is the unit of travel analysis rather than a vehicle trip. Increasingly, these models have been linked with emissions estimation tools (Johnston and De La Barra, 2000; Rakha and Ahn, 2004; Shiftan and Suhrbier, 2002; Steadman et al., 2004), but none of these models have integrated the emissions and assignment components in such a way that the two models provide feedback that influences travel behavior. Few of these studies have attempted to use those tools to provide policy analysis or insights from the integration of a travel demand model and emissions model. For example, Hatzopoulou et al. (2007) pair an activity based model with MOBILE6.2 in the Greater Toronto Area. While their work adds a finer-grain of behavioral detail to the emissions model input, the integration remains a post-processor. That is, emissions are calculated at the end of the highway assignment process, with no feedback to the behavioral components of the model.

A primary location for emissions feedback in a travel demand model is the highway assignment algorithm. Though emissions are effected by a variety of travel decisions, route choice has a significant impact on emissions production. Tzeng and Chen (1993) found incorporating pollutant estimations into a travel assignment model, emissions were significantly reduced compared to standard assignment routines. Ahn and Rakha (2008) investigated emissions from route choice behavior in Northern Virginia, finding

route selection plays a large role in emissions. This led the researchers to the conclusion that user equilibrium (UE) or social optimum (SO) assignments with emissions minimization formulations produced optimal results.

This paper develops analytical models to examine the effectiveness of four demand side emissions reduction mechanisms. Three of the models analyze pricing tools including direct emissions pricing. The fourth model examines the effectiveness of reducing excess commuting by optimizing local commute to work patterns. Results from these models provide an indication of the level of emissions reductions possible in a best-case scenario; acknowledging that real world reductions will likely fall short of these modeled results. This analysis provides direction for planners and policy makers on the most effective mechanisms to pursue. The models and their methodologies are described in detail in the following section.

3. MODELING FRAMEWORK

To estimate emissions, a number of planning agencies apply pollutant specific rates and congested speeds from traffic assignment model results into a post processor. Network traffic flow is determined by solving a traffic assignment problem under the condition of user equilibrium (Sheffi, 1985). The fundamental aim of the traffic assignment process is to reproduce in the transportation demand model, the pattern of vehicular trips/personal trips observed on the actual highway network by employing behavioral models and assigning to the network an estimated demand for travel (represented by the trip matrix, or matrices). The model state that reproduces observed network conditions is called the Base-case in this paper. The Base-case user equilibrium formulation is provided in the Appendix, Part One. The algorithm developed in this paper relies on inputs commonly found in demand models, which includes the characteristics of the transportation networks (highway and transit), socio-economic and other inputs needed for the trip generation, destination choice, mode choice and traffic assignment programs. The traffic assignment module is solved with a Frank-Wolf (FW) algorithm. When the highway assignment program is complete, the emissions model calculates the Base-case emissions. The algorithm then

diverges to run the models for (1) first-best emission pricing, (2) second-best emission pricing, (3) gas tax, and (4) commute efficiency. If the shortest emissions path is selected, the model performs the first iteration of the modified assignment with the emissions minimization function. The emission functions are shown in equations 7, 8, 9 and 16 in the Appendix. After the first iteration the model checks for convergence then recalculates emissions based on that iteration's assignment results. The iterative procedure between assignment and emissions algorithms continues until a pre-determined convergence criterion is met. The complete model structure is implemented within the Cube Voyager transportation planning software package (Citilabs, 2013).

3.1 First-best Emission Pricing

The presence of each driver on a particular route results in an additional reduction of traffic flow. This reduction in traffic flow results in real costs to other drivers in terms of time. The flow reduction also lowers the operational efficiency of the highway, producing more emissions from each vehicle. First-best emissions pricing charges all users for the marginal social cost of the emissions that result form their use of a facility. The cost of travel on a link is set at a price such that the improvement in flow (from drivers selecting other destinations, modes and routes) and the revenue from the toll (from drivers that decide to continue on the priced link) will be sufficient to offset the societal costs of the trip.

In this paper a set cost per gram of emissions is charged to each user on the network. The total cost that accrues to a driver is based on the amount of emissions that a particular vehicle adds to the output of emissions on the link and the additional emissions that result from other driver's reduced efficiency. We add an additional layer of complexity by setting an emissions cap, where each driver is only charged if driving on a road that exceeds a pre-determined emissions threshold. This method allows planners to set a goal for emission reduction; only instituting a charge for those links that fail to meet the target. The emission charge used in this research is based on an extensive literature review by Nordhaus and Boyer (2003) showing a lower bound of marginal CO_{2e} emission costs of $20 per ton. Tol (2005) provides an upper bound with a marginal cost of CO_{2e} emissions at $50 per ton. We add to this the

estimated cost of Volatile Organic Compounds (VOC) and Nitrous Oxide (NO_x) to arrive at a total emission charge of $60 per ton. The average passenger vehicle – consuming gasoline - has a fuel economy of 24.5 mpg that consume fuel at the rate of 0.008887 metric tons per gallon or .00036 metric tons per mile the total cost of emissions is $0.02 per mile. This rate is highly variable. In some cases the charge is lower than $.0005 per mile if a facility is at or near free flow conditions and in other cases where the corridor is heavily congested, the charge is much more than $0.02 per mile.

The emission price for each link is updated at the end of every traffic assignment iteration based on the emissions produced as a result of that traffic flow at that iteration over the predetermined cap. Changes in the results of each assignment are reflected in the travel cost faced by each user which, like travel time, will vary between iterations. The mathematical formulation of the emissions charge and threshold are provided in Appendix, Part Two.

3.2 Second-best Emission Pricing (VMT Tax)

An alternative to first-best pricing is to impose fees based on the number of miles driven on a roadway, commonly called a vehicle miles travelled (VMT) tax. This is a second-best solution because users will be charged a flat rate based strictly on the average social cost of their trip. An average cost mechanism is differentiated from marginal cost because a driver that causes no reduction in traffic flow will pay the same per mile charge as a user taking a more congested route, reducing the flow. This pricing method is less economically efficient as it still has the potential to result in unpaid externalities, but is mathematically less complex than a first-best mechanism. In this paper, we select a VMT tax of $0.005 per mile for LDVs and $0.01 for heavy-duty vehicles. This is consistent with much of the literature (Carroll-Larson and Caplan, 2009). The formulation of the second-best toll is provided in Appendix, Part Three.

3.3 Second-best Gas Tax

A gas tax is another way to impose a higher cost on highway travel. Like a VMT tax, it does not charge users for the marginal cost of externalities. However, the gas tax does conceptually differ from a VMT tax in three ways. First, the amount of gas consumed and subsequently the amount of taxes paid varies depending on the type of vehicle a road user drives. Thus, the driver has more control over total travel cost, subject to income constraints. Second, a gas tax is charged upfront (before a trip is taken) and generally hidden within the price of fuel. Users are less likely to link driving behavior to added fuel cost (Li et al., 2012). Third, while drivers do not closely link gas taxes to specific travel behavior such as trip timing and route selection, studies have shown that drivers typically have a higher consumption elasticity for gas prices than for road charges. This difference in elasticity is likely due to a difference in substitution options (Parry and Small, 2005). This paper uses a gas tax of $0.50 per gallon, assuming a fleet-wide average fuel efficiency of 24.5 mile per gallon, which is consistent with literature (Parry et al., 2007). Using average 24.5 mile per gallon efficiency, per mile cost of the gas tax increase is very similar to the emissions charge, a rate of $0.02 per mile. The gas tax formulation is provided Appendix, Part Four.

3.4 Commute Efficiency

Excess commuting is the amount of travel in a given area that exceeds the absolute minimum, which is derived by finding the closest suitable employment for local residents. The excess or wasteful commute (Hamilton and Röell, 1982; Hamilton, 1989) is the additional journey-to-work travel represented by the difference between the actual average commute and the smallest possible average commute, given the spatial configuration of workplace and residential locations (Ma and Banister, 2006). This extra travel leads to additional mobile emissions. A number of studies on excess commuting directly relate imbalances the location of jobs and housing to the production of wasteful travel (Layman and Horner, 2010; Loo and Chow, 2011a, 2011b). Much of the current research asserts that a higher jobs housing balance will lead to reductions in excess commuting.

In this paper we operationalize this approach by first parsing the study area into multiple zones (Niedzielski, 2006). We then determine the current number of local (zonal) jobs and the number of suitable workers within and around a zone based on job type and household income. An optimization problem is then set-up to find the minimum distance workers would have to commute if commute patterns reflected many more local jobs being filled by local residents, rather than a simulation of current observed commute patterns. Past research that takes a similar approach asserts that locations with jobs housing balances near parity, experience larger reductions in both VMT and emissions (Scott et al., 1997). This is a result of resident workers being better matched with local jobs, reducing the number of workers that commute outside of their local area.

We examine this hypothesis by inserting our newly optimized zone-to-zone travel patterns into the transportation model in substitution of the destination choice module. The method of setting up the commute efficiency optimization problem is discussed in Appendix, Part Five. It should be noted that the resulting change in job location and commuting patterns simulate an optimal condition that does not necessarily represent what is achievable in a real world context. In many cases, residential preference and job location prevent the most efficient spatial organization. However, this scenario is instructive in showing both the magnitude of excess commuting in a region and the benefits of policies that attempt to reduce the waste.

3.5 Emissions Estimation

The emissions module uses rates calculated by the EPA's MOVES2010a mobile emissions model. The module developed for this research uses a national data set of vehicle fleet age distributions, specific fuel formulation and fuel supply. County level data is used to calculate link level (road segment) emissions on the network and is then aggregated to obtain a model-wide emissions summary. Local meteorology data is used to calculate the temperatures at which vehicles are started and operated, an important factor in estimating the level of emissions produced by motor vehicles. The MOVES model provides emissions rates for vehicles categorized into 13 source types. The distribution of vehicles (source types) in the study

area was determined using data from the Maryland State Highway Administration (MSHA) and Department of Motor Vehicles. The emissions model takes congested roadway data from the transportation model and sorts the speed of each link into speed bins that are appended the highway link. Differential emissions rates are applied to various modeled urban and rural road types including freeways and highways with limited access, unrestricted highways without controlled access, arterials and minor local roads. The emissions model categorizes each of the travel demand model's network links into MOVES defined road types so that emission rates at the link level reflect the unique parameters of each area and facility type. Finally, the emissions model uses two fuel formulations, gasoline and diesel. Gasoline emission rates are applied to light duty vehicles in the model while diesel emission rates are applied to heavy-duty vehicles.

4. CASE STUDY

The proposed framework is applied to Montgomery County, Maryland. Montgomery is the most populous county in the state with one million residents, 400,000 households, and 600,000 jobs. The county's political boundary and transportation network is presented in Figure 1. The county contains parts of the most heavily travelled roadways in the Washington DC-Baltimore region. The county has an extensive highway network with parts of the Capital Beltway (Interstate-495), Interstate-270 and a portion of US Highway 29, a major route connecting the Washington and Baltimore beltways. The county also has an extensive transit network served by the MARC commuter rail, Washington Metropolitan Transit Authority (WMTA) bus service and the county's own "RIDE-ON" bus service.

The simulated version of Montgomery County's network used in this study has a total of 4,302 highway links representing over 3,630 lane miles, 73 bus routes and two rail lines. We divided Montgomery County into 159 modeling zones based on aggregations of census blocks with boundaries that follow the road geometry. Within the county there are 36 municipalities, shown in Figure 2.

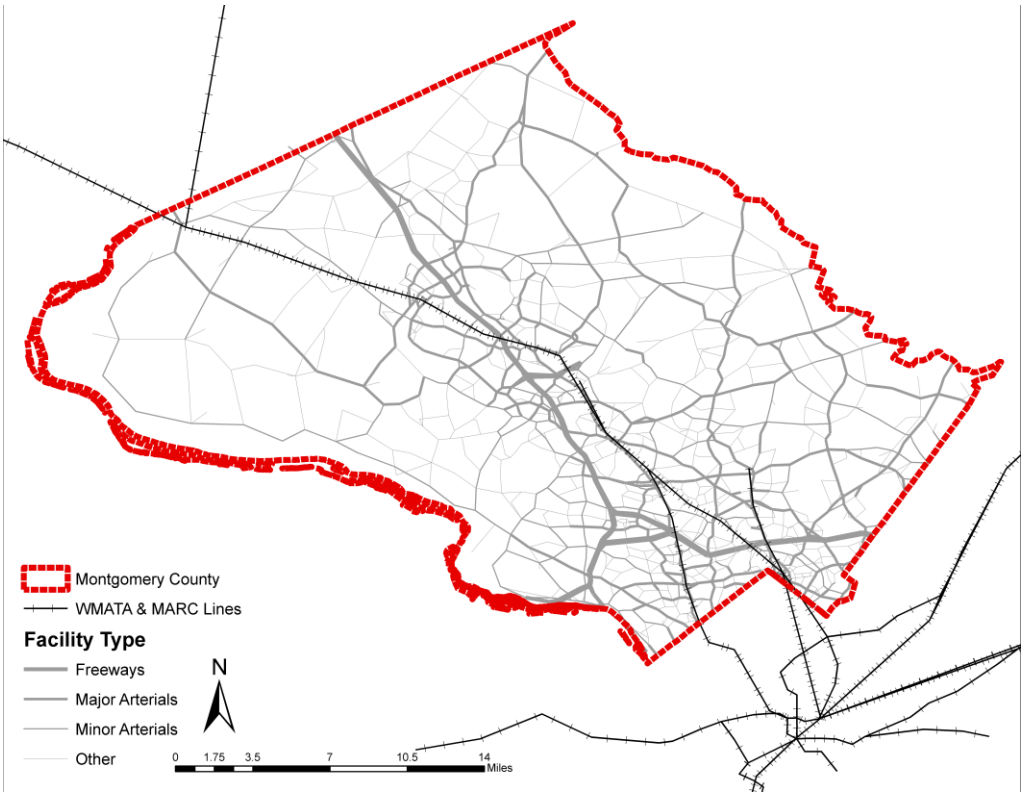

Fig. 1 Montgomery County Transportation Network

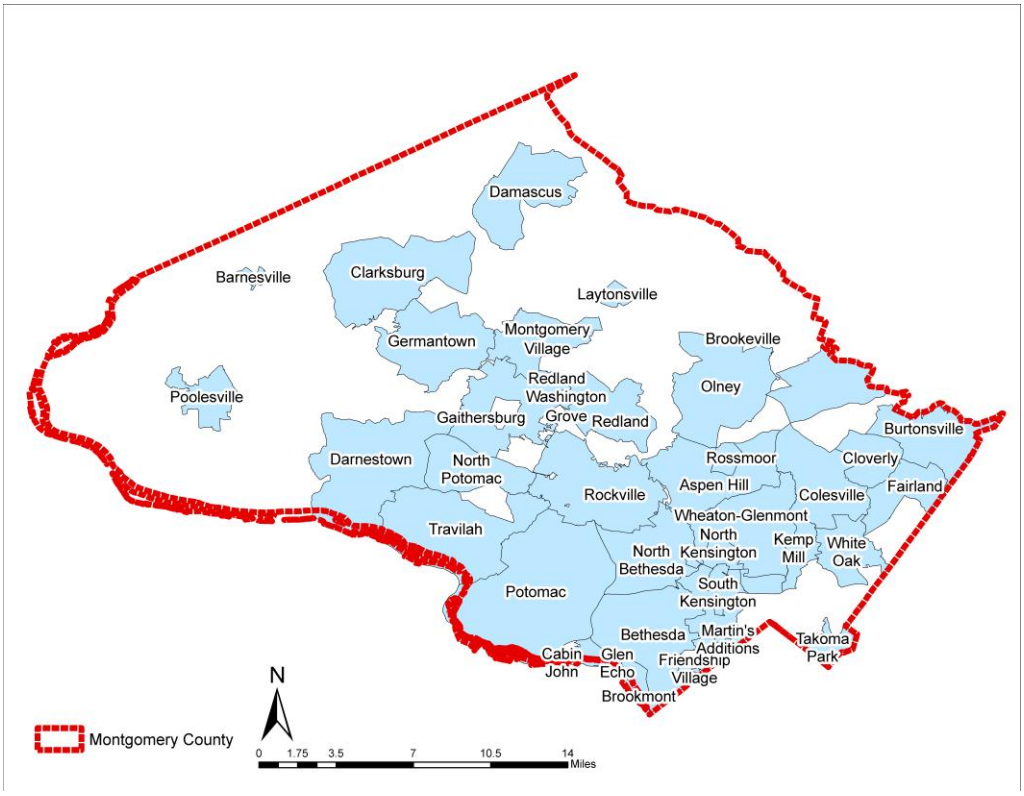

Fig. 2 Montgomery County Municipalities.

Montgomery County presents an excellent case study, as it is representative of many large and small jurisdictions interested in emissions reductions. As Figure 1 shows the county has a complex multimodal transit system with a host of transportation issues. Within the county is an array of jurisdictions composed of dense business districts, highly industrialized zones, low-density suburbs and rural agricultural lands. The results of policies implemented in the county are similar to the effects felt by many other jurisdictions.

5. RESULTS

Results are reported for the Base-case and four proposed models during a typical weekday AM peak period (6:30am - 9:30am). Table 1 provides an inventory of three types of vehicle emissions, (NO_x), (VOC) and (CO_{2e}) along with estimated changes in total system emissions (TSE), total system travel time (TST), vehicle miles traveled (VMT) and average link speed. Each emission types follow specific pattern in each model, either increasing or decreasing, though the magnitude of change for each differs. For the first two models, based on direct pricing, the largest rate of emission reduction is achieved for NOx. Road speed is a primary influence on total NO_x emissions. As a result, NO_x emissions are greatest on highway facilities and shrink with a reduction of facility vehicle capacity. In the other two models, which use either an indirect charge (gas tax) or non-network mechanisms, the largest rate reduction was for VOCs. VOCs occur in large part from evaporation either while the engine cools down or as it travels at a low speed in hot weather, called a 'running loss'. The incentive in the first two models is for drivers to seek alternative routes, away from freeways. The route switching behavior, moving more trips to lower capacity facility types, reduces overall speed, and with it NO_x. While other emissions decrease, NO_x has the greatest relationship with speed and is therefore the most sensitive pollutant to route choice. The last two scenarios result in reduced travel on lower speed facilities to a greater extent than the first two scenarios. As a result, few evaporative emissions occur on these routes and VOC reductions are more pronounced.

TABLE 1 Emissions and Highway Results by Facility Type

		Absolute Values in Millions						Avg.	Percent Change					
Model (1)	**Facility Type (2)**	**NOX (gm) (3)**	**VOC (gm) (4)**	**CO2EQ (gm) (5)**	**TSE (gm) (6)**	**TST (min) (7)**	**VMT (8)**	**Speed (MPH) (9)**	**NOX (gm) (10)**	**VOC (gm) (11)**	**CO2EQ (gm) (12)**	**TSE (gm) (13)**	**TST (min) (14)**	**VMT (15)**
Base-case	**Freeways**	4.9	0.4	904.2	909.5	0.0	1.6	44.9						
	Major Arterials	2.9	0.5	663.9	667.2	0.1	1.5	26.2						
	Minor Arterials	0.6	0.1	156.5	157.2	0.0	0.5	25.5						
	Other	1.0	0.2	274.3	275.5	0.0	0.8	19.3						
	TOTAL	***9.3***	***1.2***	***1,998.9***	***2,009.4***	***0.2***	***4.4***	***29.0***						
Model-1 (First Best)	**Freeways**	2.2	0.2	406.9	409.3	0.0	0.8	50.5	-55.2	-55.1	-55.0	-55.0	-67.1	-54.1
	Major Arterials	2.5	0.4	575.4	578.3	0.1	1.3	24.3	-12.7	-13.7	-13.3	-13.3	88.9	-14.0
	Minor Arterials	0.6	0.1	181.4	182.2	0.1	0.6	23.2	17.1	16.3	15.9	15.9	244.3	13.6
	Other	1.4	0.3	354.7	356.3	0.2	0.9	18.2	39.7	35.0	29.3	29.4	339.8	12.8
	TOTAL	***6.7***	***1.0***	***1,518.4***	***1,526.2***	***0.4***	***3.5***	***29.1***	***-27.8***	***-17.2***	***-24.0***	***-24.0***	***123.6***	***-21.0***
Model-2 (VMT tax)	**Freeways**	4.2	0.4	780.5	785.1	0.0	1.4	47.3	-13.3	-13.8	-13.7	-13.7	-27.4	-15.0
	Major Arterials	2.6	0.4	597.0	600.0	0.1	1.3	26.6	-9.5	-10.0	-10.1	-10.1	-13.3	-11.5
	Minor Arterials	0.5	0.1	143.1	143.7	0.0	0.5	25.6	-8.2	-8.2	-8.6	-8.6	-8.2	-9.7
	Other	0.9	0.2	267.3	268.4	0.0	0.8	19.3	-3.3	-2.3	-2.6	-2.6	-1.3	-2.1
	TOTAL	***8.3***	***1.1***	***1,787.8***	***1,797.3***	***0.1***	***4.0***	***29.7***	***-10.8***	***-9.9***	***-10.6***	***-10.6***	***-13.8***	***-11.0***
Model-3 (Gas Tax)	**Freeways**	4.7	0.4	869.7	874.9	0.0	1.6	45.9	-3.8	-3.8	-3.8	-3.8	-10.5	-4.1
	Major Arterials	2.7	0.4	627.1	630.3	0.1	1.4	26.5	-5.4	-5.5	-5.5	-5.5	-7.5	-5.9
	Minor Arterials	0.5	0.1	146.8	147.4	0.0	0.5	25.6	-6.1	-6.1	-6.2	-6.2	-10.0	-6.7
	Other	0.9	0.2	266.2	267.3	0.0	0.8	19.4	-3.3	-3.0	-3.0	-3.0	-3.1	-2.6
	TOTAL	***8.9***	***1.2***	***1,909.8***	***1,919.9***	***0.2***	***4.2***	***29.3***	***-4.4***	***-4.5***	***-4.5***	***-4.5***	***-7.7***	***-4.7***
Model-4 (Commute Efficiency)	**Freeways**	4.8	0.4	884.5	889.8	0.0	1.6	45.0	-2.2	-2.3	-2.2	-2.2	-3.7	-2.0
	Major Arterials	2.6	0.4	603.6	606.7	0.1	1.4	26.4	-9.2	-9.2	-9.1	-9.1	-9.7	-8.6
	Minor Arterials	0.5	0.1	141.1	141.7	0.0	0.5	25.6	-10.0	-10.0	-9.9	-9.9	-12.0	-9.4
	Other	0.9	0.2	265.2	266.3	0.0	0.8	19.4	-4.3	-3.5	-3.3	-3.3	-5.0	-2.0
	TOTAL	***8.9***	***1.1***	***1,894.4***	***1,904.4***	***0.2***	***4.2***	***29.1***	***-5.1***	***-5.9***	***-5.2***	***-5.2***	***-7.3***	***-5.1***

Each of the models reduces aggregate emissions, but by significantly different amounts. In Model-1, using a direct emissions charge provides the greatest emissions reduction, with a decrease of over 24% compared to the Base-case. This reduction comes at the cost of higher TST, particularly on minor arterials and other roads. This results from drivers seeking lower cost routes with less volume and congestion (consequently fewer emissions). The substantial shift from freeway and major arterial travel to minor arterial and local roads shown by the change in VMT is evidence of this cost reduction behavior. The increase in travel time and decrease in emissions represents a trade-off for those that value the dollar cost of travel over travel time savings.

The VMT tax, which also charges road users for specific link usage, results in the second biggest decrease in emissions (-10.56%). Drivers are not able to realize travel cost savings by simply taking less travelled routes, since road users are priced for every mile of driving within the study area regardless of the type of facility they use. There is more incentive for drivers to change destinations or modes if the travel cost exceeds their driving utility. Thus, the VMT tax provides both a travel time (-13.79%) reduction and an emissions reduction.

The gas tax results in less of an emissions reduction; an expected result both in a modeling and real world context. This is an effect of the previously noted buried nature of the gas tax. A modest 4.46% reduction is achieved for emissions, 7.68% reduction in travel time and 4.75% reduction in VMT for this policy. The gas is the least effective of the four emission reduction mechanisms in reducing any of the aggregated emissions.

The excess commute model optimizes travel by better matching local residents with local jobs. This maximizes intra-zonal travel where supported by a high jobs housing balance, reducing the amount of time a resident spends driving to work. Optimizing job location and commute patterns reduces total emissions by more than 5%. Aggregate travel time (-7.34%) and VMT (-5.11%) is also reduced with this mechanism. Commute optimization reduces these metrics much more for major and minor arterials and other facility types, particularly compared to other mechanisms. This is a result of the minimization of travel time between households and jobs. If the jobs housing balance does not support a fully internal

commute, zone-to-zone commute travel is based on the least travel time. This typically means travel to zones served by freeways is favored, increasing travel on these facilities and reducing travel on all other facility types.

Table 2 Excess commuting for workers residing in Montgomery County

City	Jobs per Resident Worker	Commute type		
		Required (min)	Base (min)	Excess (%)
Ashton-Sandy Spring	0.43	6.46	22.93	71.84
Aspen Hill	0.32	8.44	16.74	49.59
Barnesville	0.62	9.40	28.01	66.44
Bethesda	3.30	3.36	10.74	68.73
Brookmont	1.50	3.20	15.10	78.82
Burtonsville	2.19	4.29	21.46	80.00
Cabin John	0.34	5.14	13.89	62.98
Calverton	1.25	2.81	14.06	80.01
Chevy Chase	0.66	3.45	10.40	66.85
Clarksburg	0.39	7.96	19.56	59.31
Cloverly	0.30	12.79	19.18	33.32
Colesville	0.36	11.25	17.11	34.27
Damascus	0.32	10.92	22.35	51.16
Darnestown	0.49	14.59	18.23	19.98
Fairland	0.43	7.85	17.85	56.03
Forest Glen	1.12	4.73	11.48	58.82
Gaithersburg	1.58	5.64	12.48	54.83
Germantown	0.47	13.11	14.88	11.86
Kemp Mill	0.41	5.73	14.20	59.63
Laytonsville	0.14	10.98	22.76	51.75
Montgomery Village	0.59	4.20	13.79	69.54
North Bethesda	2.27	4.23	11.11	61.89
North Potomac	0.11	7.18	15.59	53.94
Olney	0.32	13.47	19.71	31.66
Poolesville	0.21	27.64	33.72	18.02
Potomac	0.72	6.87	14.74	53.41
Redland	0.65	7.03	16.28	56.79
Rockville	1.56	4.65	12.67	63.34
Rossmoor	0.38	9.47	17.66	46.39
Silver Spring	1.35	4.40	11.83	62.84
Somerset	1.94	2.02	11.96	83.12
South Kensington	0.82	4.43	10.16	56.44
Takoma Park	0.64	3.58	14.08	74.55
Travilah	0.36	11.70	17.59	33.49
Wheaton-Glenmont	0.73	6.49	13.29	51.17
White Oak	1.08	11.16	14.51	23.08
County	1	7.79	16.45	54.33

Table 2 shows the impact of excess commuting problem on travel time. The study area, Montgomery County, was divided into 159 zones, represented by 36 municipalities (see Figure 2). Poolesville has the highest commute time among the cities. The city is located the farthest distance from the main urbanized areas of the county. Poolesville also has one of the lowest jobs to resident worker ratios (0.21 jobs per resident), which means it is difficult to achieve significant commute efficiency (just 18%). On the other hand, the Township of Somerset has the highest excess commute, over 83%. This is due to its proximity to the urban core and its high jobs to resident worker ratio with nearly two jobs for every worker. The high jobs to housing ratio indicate many more local jobs can be filled by local residents, significantly reducing total commute travel. On average, workers residing within the study area commute for 16.45 minutes in peak hour traffic. The excess commute results indicate that the average required commute is 7.79 minutes. As a result, an average of 54.33% of the current commute is wasteful.

At the link level, each model has a unique impact on the emissions along every network link. Figure 3 shows the results on emissions for links grouped by peak capacity. In the Base-case, emissions resulted in smooth growth within each capacity class. For Model-1 the links ordered in the same way as the Base-case show the significant impact on various links because of emission pricing. The peak emissions for all links are much lower than in the Base-case. For Model-2 the results show evidence of link emission reductions but the overall trend is similar to the Base-case. The last two models are very similar to the Base-case, but with lower emission rates for some links, particularly with a 1,200 vehicle capacity and lower peaks for some of the 1,800 to 3,000 vehicle capacity links.

Figure 4 is a three-dimensional representation of emissions for all four models and the Base-case, measured as total aggregate emissions from each link. The figure shows the magnitude and scale of emissions resulting from each model. In the Base-case (first layer), emissions on the highways is apparent from the substantial peaks. Model-1 (second layer) shows the effects of a direct emissions pricing. The peaks compared to the Base-case are much smaller, but the surface becomes very spread out, indicating more emissions are being produced on arterials and local roads. Model-2 (third layer) is a mixture of

Base-case peaks and the bumpy Model-1 non-highway surface; a result of drivers' inability to avoid the VMT tax. The final two layers show spatially the modest improvements over the Base-case. The magnitudes of the emissions along certain segments of the highway are similar to the Base-case, but some of the emissions reductions are due to less highway driving and slightly more dispersed local road use. This is consistent with the two methodologies which both encourage less long distance travel and fewer internal-external and external-internal trips.

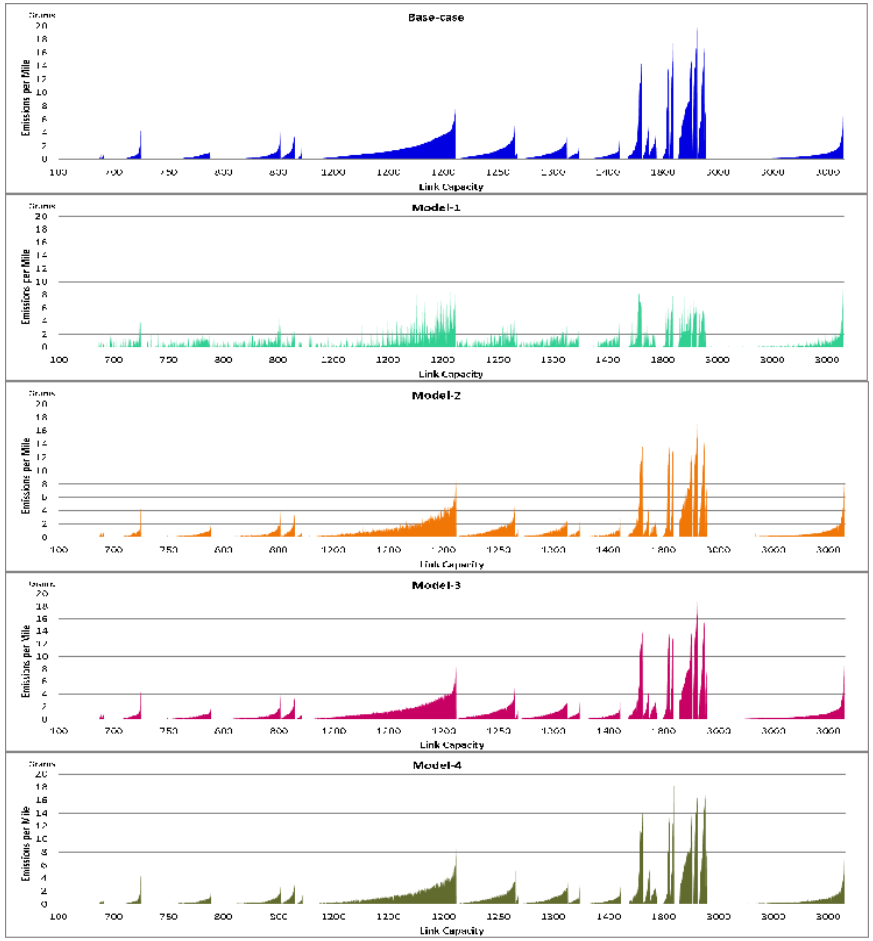

Fig. 3 Aggregate link emissions per mile grouped by link capacity

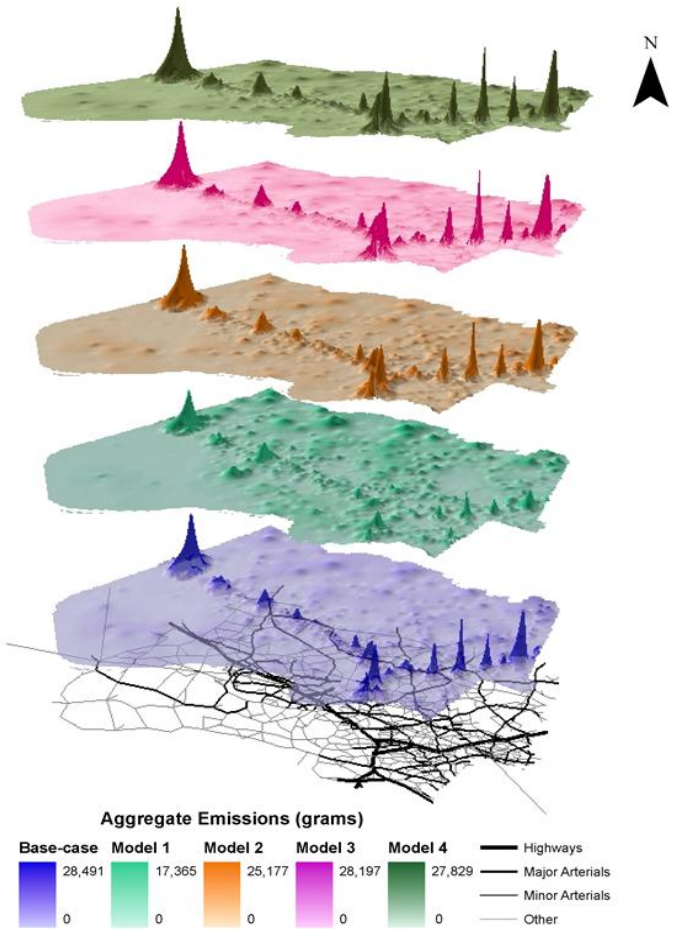

Fig. 4 Interpolated total aggregate link emissions

6. DISCUSSION AND CONCLUSION

This paper proposed an integrated travel demand and emissions model that accommodates a number of policy mechanisms. Each of the mechanisms have the potential to significantly reduce emissions. MOVES-based emission formulations are integrated with the transportation demand model, by applying factors based on link level data disaggregated by area-type, facility-type, speed, fleet composition, time of day and area-specific meteorology and fuel formulation. Four distinct emission reduction mechanisms have been examined: 1) first-best emissions pricing, 2) second-best VMT-based tax 3) second-best gas tax, and 4) commuting efficiency. Each of these policies has been independently examined using a case study in Montgomery County, Maryland. Emissions from each model are measured against each other, in a single application, to determine the most likely candidate to achieve substantial emission reductions.

Results from the four emission reduction policies suggest there is no single greatest strategy for emission reductions. Each model requires some degree of trade-off between emission reduction, travel time or amount of VMT. The best candidate, given the costs (in terms of congestion or travel time) of some policies and the limited efficacy of others, is likely one that combines feasible elements of each, with broad multijurisdictional policy implementation. A first-best pricing option is the most effective emission mechanism with significant reductions on freeways and a 24% total emission reduction. However, this must be weighed against increased travel time and emissions on lower capacity facilities, a likely high cost of implementation and difficulties to price correctly. The second-best options presents reasonable emission reductions, with an aggregate decrease ranging from 4.5% to over 10%, though the addition of any perceived tax, either road or gas, might make these mechanisms politically unfavorable. The non-network based commute efficiency policy is likely to suffer less from limited jurisdictional implementation, though enforcement may be difficult and administration costs could potentially outweigh the benefits. While the four mechanisms do achieve their stated goals of emission reductions, there are many factors that may limit their effectiveness including jurisdictional constraints, economic inefficiency and uncertainty about future implementation. The prices selected in this paper are based on a review of

the existing literature. It should be noted that changing the price of any one of the demand-based models would significantly alter the model outcome.

This paper shows that achieving vehicle emission reductions is a complicated goal, as there are substantial uncertainties regarding road user's response to each policy. The implementation of a particular policy may result in substantial emission reductions on one facility type, but could have significantly detrimental emissions and travel time impacts on other facilities. The research leaves open the question of how large a policy area is needed and the level of regional coordination required to achieve optimal emissions reductions. Future research efforts in multiple policy analysis will benefit from the use of advanced travel demand modeling techniques that are now in the early stages of adoption by planning agencies. The models could also be extended to perform a benefit cost analysis of each emission reduction mechanism.

APPENDIX

Notation		Explanation
C_1	:	The average commute cost from the commute optimization operation
C_1	:	Average commute before optimization
C_2	:	Average commute cost after optimization
C_a	:	The capacity for link a
C_{excess}	:	The excess commute derived from commute optimization
D_{ij}^k	:	Various distance terms (linear, log, squared, cubed and square root)
e_a	:	Emission price
f_{ij}^r	:	Flow on path *r*, connecting each Origin-Destination (O-D) pair (*i-j*)
l_a	:	Distance for link a
q_{ij}	:	Demand between each Origin-Destination (O-D) pair (*i-j*)
t_a	:	Travel time for link a
$t_a(x_a)$	:	Travel cost on link a as a function of flow
t_{ij}	:	Travel cost between origin *i* and destination *j*
$u_a^I(x_a, e_a,)$	:	Travel time function for Model-1 which incorporates emission pricing term e_a
$u_a^{II}(x_a, \theta_a,)$	:	Travel time function for Model-1 which incorporates VMT tax term θ_a
$u_a^{III}(x_a, \sigma)$	:	Travel time function for Model-1 which incorporates gas tax term σ
u_a	:	User cost for link a
u_{ij}^c	:	Least cost path between O-D pairs *i-j*
x_a	:	Flow for link a
α_a	:	Constant, varying by facility type (BPR function)
β_a	:	Constant, varying by facility type (BPR function)
β^k	:	Weights for each term in the size variable (S_j)
γ^c	:	Value of time (VOT) for user class c
$\delta_{a,ij}^r$	:	Flow on link a, a subset of path *r*, connecting each Origin-Destination (O-D) pair (*i-j*)
τ_a	:	Toll value for link a
Φ_a	:	Emissions cap for each link a
ϕ_a	:	Total emissions for link a
c	:	User class
d_{ij}	:	The number of commuter trips between *i* and *j*
n	:	Assignment iteration number
T	:	The total number of commuters
t_o	:	Free flow time on link a
φ	:	Emissions charge per gram of emissions, in cents
ω	:	A positive constant (exponential demand function)

Part One: Base-case

This principle is based on the fact that individuals choose a route in order to minimize their travel time or travel cost and such a behavior on the individual level creates an equilibrium at the system (or network) level over a long period of time (Sheffi 1984). Simply, for each origin-destination (O-D) demand pair, the travel-cost/travel-time on all used routes of the road network should be equal.

$$Minimize \sum_{a} \int_{0}^{x_a} (t_a(x_a)) \tag{1}$$

Subject to:

$$\sum_{r} f_{ij}^{r} = q_{ij} \tag{2}$$

$$x_a = \sum_{i} \sum_{j} \sum_{r} f_{ij}^{r} \, \delta_{a,ij}^{r} \tag{3}$$

$$f_{ij}^{r}, q_{ij}^{r} \geq 0 \tag{4}$$

Equation (1) represents that at equilibrium the network will satisfy the User Equilibrium (UE) condition, i.e. travel time on all the used routes connecting any given *i-j* pair will be equal. The term, t_a, is the travel time for link *a*, which is a function of link flow x_a. Equation (2) is a flow conservation constraint to ensure that flow on all paths *r*, connecting each Origin-Destination (O-D) pair (*i-j*) is equal to the corresponding demand. In other words, all O-D trips must be assigned to the network. Equation (3) represents the definitional relationship of link flow from path flows. Equation (4) is a non-negativity constraint for flow and demand. The travel time function $t_a(.)$ is specific to a given link '*a*' and the most widely used model is the Bureau of Public Roads (BPR) function given by

$$t_a(x_a) = t_o \left(1 + \alpha_a \left(\frac{x_a}{C_a}\right)\right)^{\beta_a} \tag{5}$$

where *to*(.) is free flow time on link '*a*', and α_a and β_a are constants (and vary by facility type). C_a is the capacity for link a. In the base model the objective is minimization of total system travel time.

Part Two: First-Best Emission Pricing

The emissions cap for each link is:

$$\Phi_a = \left[\frac{1}{N} \sum_{a=1}^{N} \left(\frac{\phi_a}{l_a}\right)\right] * l_a \tag{6}$$

where ϕ_a is the total emissions for link a calculated for each link in the base model and l_a is the link distance. Once the cap is determined, the emission price (e_a) can be incorporated into the travel demand model. The emission price can be converted to travel time units with appropriate factor (γ^c) representing

VOT in monetary terms as cents per minute for travellers of five income categories c. The revised user cost function for link based emission is

$$u_a^I(x_a, e_a) = t_a(x_a) + \frac{\varphi e_a(x_a)}{\gamma^c} \tag{7}$$

where $u_a^I(x_a, e_a,)$ is the travel cost function for Model-1, which incorporates emission pricing term e_a. The objective function for Model-1 is similar to base case with the exception that the third term from equation (7) ($\frac{\varphi e_a(x_a)}{\gamma^c}$) is added to equation (1) which is the total emissions e produced on link a, which is a function of link flow x_a multiplied by charge per gram of emissions, φ.

Part 3: Second-Best Emission Pricing (VMT Tax)

Analytically, the user cost function can be stated as the following to incorporate the VMT based tax.

$$u_a^{II}(x_a, \theta_a) = t_a(x_a) + \frac{\theta_a l_a}{\gamma^c} \tag{8}$$

where, θ_a is the VMT tax in \$/mile for link a, $l_a$is the link length in miles, and γ^c is the VOT in \$/hour. In traffic assignment procedure, the user cost shown in equation (8) can be used in equation (1). The advantage of VMT based tax is to encourage travelers to use transit as an alternate mode if the tax appears too onerous. Equation (8) refers to VMT based tax associated with value of time (VOT).

Part Four: Gas Tax

The effect of gas price on user behavior can be implemented as follows:

$$u_a^{III}(x_a, \sigma) = t_a(x_a) + \frac{\sigma l_a}{\gamma^c \vartheta} \tag{9}$$

where, σ is the gas price in dollars per mile (as a ration of dollars per gallon and fleet-wide efficiency of 24.5 mpg), $l_a$is the link length in miles, γ^c is the VOT in \$/hr, and ϑ is the automobile gasoline efficiency in miles per gallon. Auto Operating Cost (AOC) is another component which is considered in the mode choice model (please see equation 19). A higher gas price will result in a higher AOC and therefore will make auto travel more expensive.

Part Five: Commute Efficiency

$$Minimize \sum_i \sum_j u_{ij} d_{ij} \tag{10}$$

Where, $u_{ij} = t_{ij}$ and t_{ij} is the travel cost between origin i and destination j and d_{ij} is the number of commuter trips between i and j. The constraints for the optimization problem are subject to

$$\sum_j d_{ij} = O_i \tag{11}$$

$$\sum_i d_{ij} = D_j \tag{12}$$

$$d_{ij} \geq 0 \tag{13}$$

The average commute cost from the optimization can be interpreted as

$$C_1 = \frac{1}{T} \sum_i \sum_j u_{ij} d_{ij}^o \tag{14}$$

Where, T is the total number of commuters, and d_{ij}^o is the optimal number of commuters between *i-j* pair. Before optimization, the average commuter cost was:

$$C_2 = \frac{1}{T} \sum_i \sum_j u_{ij} d_{ij} \tag{15}$$

The excess commute can be defined as the percentage difference between cost before and after optimization. This can be represented as

$$C_{excess} = \frac{(C_2 - C_1) * 100}{C_1} \tag{16}$$

The excess commute can be considered as a dis-benefit from the planner's viewpoint.

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